Week 5: Hypothesis Design and Power Calculations

Empirical Research Methods

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Learning Objectives

- Understand the role of hypothesis testing in research design
- Learn to formulate null and alternative hypotheses
- Understand Type I and Type II errors
- Calculate statistical power and Minimum Detectable Effects
- Determine appropriate sample sizes for your study

Hypothesis Testing Framework

The Foundation of Inference

Null Hypothesis (H_0)

What you are attempting to reject. The default assumption: there is no effect or no difference.

$$H_0 : \tau = 0 \quad \text{(treatment has no effect)}$$

Alternative Hypothesis (H_1 or H_a)

The claim we're testing: there is an effect or difference.

$$H_1 : \tau \neq 0 \quad \text{(two-sided test)}$$

or

$$H_1 : \tau > 0 \quad \text{(one-sided test)}$$

Hypotheses Construction I

See Bishop and Thompson 2023 chapter 3 for additional advice

Starting from an intervention you want to test (i.e. an informative pamphlet on energy consumption tips for households), identify:

- Which outcomes your intervention will change.
- How these outcomes will be measured.
- How will you model the manner in which your outcome metrics are impacted by the changes made in the intervention.

For example, if you want to test an information intervention's impact on energy consumption, this would be measured with kWh, and the intervention would take the form of a binary variable. What would this make the hypotheses?

Hypotheses Construction II

See Bishop and Thompson 2023 chapter 3 for additional advice

Starting from a variable/metric you want to change a certain way (i.e. energy consumption), outline:

- How can this metric potentially be influenced.
- How can the variables that influence the metric of interest be changed (in an experiment) or vary naturally (for an observational study).
- A model that connects the independent variables to the metric of interest.

Example question: What variables can we examine for emissions resulting from procurement at UZH and where could we look to influence them?

One-Sided vs. Two-Sided Tests

Choosing the Right Alternative

Two-Sided Test

$$H_0 : \tau = 0 \quad \text{vs.} \quad H_1 : \tau \neq 0$$

Tests whether effect exists in either direction. More conservative.

One-Sided Test

$$H_0 : \tau \leq 0 \quad \text{vs.} \quad H_1 : \tau > 0$$

Tests whether effect exists in a specific direction. More powerful but requires justification.

When to use one-sided:

- Theory strongly predicts direction
- Negative effects are not plausible
- Pre-registered before seeing data

Type I and Type II Errors

Understanding Statistical Mistakes

	H_0 **True**	H_0 **False**
Reject H_0	**Type I Error** (α)	**Correct** (Power)
Fail to Reject H_0	**Correct** $(1-\alpha)$	**Type II Error** (β)

- **Type I Error (α):** False positive (reject true null)
- **Type II Error (β):** False negative (fail to reject false null)
- **Power ($1-\beta$):** Probability of detecting true effect under a given alternative hypothesis

Understanding Statistical Power

The Probability of Detecting an Effect

Definition

Power is the probability of rejecting the null hypothesis when it is false.

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 | H_A \text{ is true})$$

Standard Practice

Aim for 80% power ($\beta = 0.20$) as a convention in empirical research.

Interpretation: If we repeat the study 100 times, we expect to detect the true effect 80 times.

See chapter 10 of Bishop and Thompson 2023 and J-PAL's Power page for more details on power calculations.

Power Calculations: Two-Sample T-Test Example

See Athey and Imbens 2017 for one detailed derivation

Intuition: Knowing the test statistic distribution under the null is a student t-distribution and supposing the alternative hypothesis is true (i.e. a *specific* value that satisfies the construction of the alternative), you can solve for the power of your experiment.

$$\text{Power} = 1 - \Phi(t_{\alpha/2} - \frac{\theta}{\sigma/\sqrt{n}})$$

where:

- $t_{\alpha/2}$ = critical value for significance level
- θ = value of parameter of interest under alternative hypothesis
- σ = standard deviation of population
- n = sample size per group

Factors Affecting Power

What Determines Your Ability to Detect Effects?

- **Effect Size**: Larger effects are easier to detect. Specify the Minimum Detectable Effect.
- **Sample Size**: Larger samples provide more power. This is what we calculate in power analysis.
- **Significance Level (α)**: Lower α (e.g., 0.01 vs. 0.05) reduces power. Standard is $\alpha = 0.05$.
- **Outcome Variance**: Lower variance (more precise measurement) increases power.
- **Design**: Stratified randomization and clustering affect power.

A website for exploring impacts of experimental conditions on power: https://egap.shinyapps.io/Power*Calculator*/.

Minimum Detectable Effect (MDE)

What Effect Size Can You Detect?

Definition

The smallest effect size you can reliably detect with your study design. Can be specified in terms of standard deviations.

For a two-sample (two-sided) t-test:

$$\text{MDE} = (t_{\alpha/2} + t_{\beta}) \times \sigma \times \sqrt{\frac{2}{n}}$$

where:

- $t_{\alpha/2}$ = critical value for significance level (from normal distribution)
- t_{β} = critical value for power (from normal distribution)
- σ = standard deviation of outcome (Why might this be tricky?)
- n = sample size per group

Example: With $n = 500$, $\sigma = 1$, standard β and α, MDE $= 0.18$.

Sample Size Calculation

How Many Participants Do You Need?

The Process: Rearrange the Previous Formula

1. Specify your desired power (typically 80%)
2. Specify your significance level (typically 5%)
3. Specify the effect size you want to detect (MDE)
4. Estimate the variance of your outcome
5. Calculate required sample size

Tools:

- Statistical software: R, Python, Stata
- Online calculators: G*Power, Optimal Design
- Research papers: Often provide sample size formulas

Power Calculation Example

Energy Efficiency Program

Study Design

- Outcome: Monthly energy consumption (kWh)
- Baseline mean: 500 kWh
- Baseline SD: 150 kWh
- Desired MDE: 5% reduction
- Significance level: $\alpha = 0.05$ (two-sided)
- Power: 80%

How can we fine the number of participants we need per group?

Power Calculation Example

Energy Efficiency Program

Study Design

- Outcome: Monthly energy consumption (kWh)
- Baseline mean: 500 kWh
- Baseline SD: 150 kWh
- Desired MDE: 5% reduction = 25 kWh
- Significance level: $\alpha = 0.05$ (two-sided)
- Power: 80%

Calculation:

$$n = 2 \times \left(\frac{(1.96 + 0.84) \times 150}{25} \right)^2 \approx 565 \text{ per group}$$

Result: Need approximately 1'130 total participants.

Practical Considerations

Attrition & Non-compliance

Expect some participants to drop out or don't comply with the assignment. Account for the compliance rate and increase sample size by 10-20%.

Clustering (more next time on this)

If using cluster randomization, multiply sample size by design effect.

Multiple Comparisons

If testing multiple hypotheses, adjust significance level (e.g., Bonferroni correction).

Budget Constraints

If you can't afford the calculated sample size, reconsider your MDE or design.

Why Multiple Testing Creates Issues

If we are conducting m tests, each of these tests has their own chance of producing a false positive. As these tests are not the same, these events are not the same and stack, increasing the risk of Type I Error.

Worst Case Scenario

$$
\begin{aligned}
P(\text{Type I error in } m \text{ questions}) &= 1 - P(\text{No Type I error in all } m \text{ questions}) \\
\textit{assuming independence} \rightarrow &= 1 - P(\text{No Type I error in } Q_i)^m \\
&= 1 - (1 - \alpha)^m
\end{aligned}
$$

Multiple Testing

Methods for control (see Oehlert 2000, chapter 5.2 for more details)

Family-Wise Error Rate (FWER)
Probability of making one or more false discoveries.

When should we use this?
When the consequence of a false positive are high, such as declaring a drug works and is ready for the market.

Statistical tools to control FWER

- Bonferroni Correction
- Bonferroni-Holm

False-Discovery Rate (FDR)
The proportion of alternative hypotheses that are falsely found to be statistically significant.

When should we use this?
If the consequence of a false positive are low, such as using statistical tests to flag genes for further investigation.

Statistical tools to control FDR

- Benjamini-Hochberg
- Benjamini–Yekutieli

Hypothesis Design in Your Pre-Analysis Plan

What to Document

Your PAP Should Specify:

- **Primary hypothesis:** Clear statement of H_0 and H_1
- **One-sided or two-sided:** With justification
- **Significance level:** Typically $\alpha = 0.05$
- **Desired power:** Typically 80%
- **Minimum Detectable Effect:** What effect are you powered to detect?
- **Sample size:** How many participants do you need?
- **Power calculation details:** Assumptions about variance, design effects, etc.

Good Practices in Hypothesis Design

Ensuring Credibility

- **Pre-register:** Specify hypotheses before seeing data
- **One primary hypothesis:** Focus on your main research question
- **Secondary hypotheses:** Clearly label exploratory analyses
- **Justify MDE:** Explain why the effect size is meaningful and practical
- **Sensitivity analysis:** Test robustness to different assumptions
- **Conservative assumptions:** Use realistic estimates of variance and design effects
- **Document deviations:** If actual sample size differs from planned, explain why

Common Mistakes in Hypothesis Design

Makin and Orban de Xivry 2019

Mistake 1: Underpowered Studies

Too small sample size means you can't detect real effects. Aim for 80% power when collecting sample units.

Mistake 2: Too Many Hypotheses

Multiple testing increases false positive rate. Focus on primary hypothesis and adjust appropriately.

Mistake 3: Unrealistic Effect Sizes

Specifying very small MDEs requires huge samples. Be realistic about practical significance. Reference the literature and consider publication bias.

Mistake 4: Ignoring Design Complexity

Use randomization wisely. We will examine this next time!

Summary: Hypothesis Design and Power

Key Concepts

- Hypothesis testing requires specifying null and alternative hypotheses
- Power consideration is vital for detecting a true effect
- Sample size depends on desired power, effect size, and variance
- Practical considerations and population characteristics affect required sample size

Next Steps

- Specify your primary hypothesis
- Determine a realistic Minimum Detectable Effect
- Calculate required sample size
- Document all assumptions in your pre-analysis plan
- Pre-register your hypothesis and power calculation

Citations and Resources I

- Athey, Susan and Guido Imbens (2017). "The Econometrics of Randomized Experiments". en. In: *Handbook of Economic Field Experiments*. Vol. 1. Elsevier, pp. 73–140. ISBN: 978-0-444-63324-8. DOI: 10.1016/bs.hefe.2016.10.003. URL: https://linkinghub.elsevier.com/retrieve/pii/S2214658X16300174 (visited on 03/24/2026).
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- Gupta, Sabhya and Sarah Kopper (Aug. 2021). *Power calculations*. en. URL: https://www.povertyactionlab.org/resource/power-calculations (visited on 03/24/2026).

Citations and Resources II


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Oehlert, Gary W. (2000). *A first course in design and analysis of experiments*. en. New York: W.H. Freeman. ISBN: 978-0-7167-3510-6.